University of Cape Town

Department of Mathematics and Applied Mathematics

Mathematics II **Advanced Calculus (2AC)**

Class Test 2 **9th May 2023**

Time: 1.5 hrs **Marks available: 36** **Full Mark: 34**

Section A Full Mark: 26

In this section full answers are expected. Marks will be deducted for incomplete solutions.

1. Find the points of maximum and minimum of the function $f(x, y, z) := -x + y^2 - z^2$ over the sphere $x^2 + y^2 + z^2 = 1$.

[5]

2. Find the cartesian equation of the tangent plane to the level surface

$$\mathcal{S} := \left\{(x, y, z) \in \mathbb{R}^3\colon\ \sin(x + y) + \sin(x + z) + \sin(y + z) = 1 + \sqrt{2}\right\}$$

at the point $\left(\frac{\pi}{4}, \frac{\pi}{4}, 0\right)$.

[4]

3. Let $\mathcal{S}$ be the surface described by the vector function $\vec{r} : D = \left[0, \pi\right] \times \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \to \mathbb{R}^3$ defined by

$$\vec{r}(s, t) := (\cos(2s),\ -\sin s + t,\ 2s - \cos t) \text{ for every } (s, t) \in D\,.$$

(a) Show that $\vec{r}_s(s, t) \times \vec{r}_t(s, t) = (-\cos s \sin t - 2,\ 2\sin(2s)\sin t,\ -2\sin(2s))$ for every $(s, t) \in D$.

(b) Show that the surface $\mathcal{S}$ is regular or smooth.

(c) Write down the cartesian equation of the tangent plane to the surface $\mathcal{S}$ at the point $\left(0, -\frac{\sqrt{2}}{2}, \frac{\pi - 2}{2}\right)$.

[2, 2, 2]

4. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) := y\sin(3x) + y^2 \qquad \forall (x, y) \in \mathbb{R}^2\,.$$

(a) Find **all** the critical points of the function f.

(b) Determine the nature (i.e., point of local maximum/minimum, saddle point) of the critical points $\left(\frac{\pi}{3}, 0\right)$ and $\left(\frac{\pi}{6}, -\frac{1}{2}\right)$.

[4, 4]

5. Let $D := \{(x,y) \in \mathbb{R}^2\colon\ x^2+y^2 \geq 1, \quad 0 \leq x \leq y, \quad y \leq 1\}$.

Set up the double integral $\displaystyle\iint_D \frac{xy^2}{(x^2+y^2)^2}\, dxdy$. in polar coordinates **without evaluating it**.

[3]

Section B **Full Mark: 10**

1. The function $f(x,y) := -x^{2023} + y^{2021}$ has

(a) a local maximum at $(0,0)$; (b) a local minimum at $(0,0)$;

(c) a global minimum at $(0,0)$; (d) a saddle point at $(0,0)$;

(e) a global maximum at $(0,0)$.

[2]

2. The double integral $\displaystyle\iint_T \sin(x+y)dxdy$ over $T := \big\{(x,y) \in \mathbb{R}^2\colon\ 0 \leq x \leq \pi/2,\ x \leq y \leq \pi/2\big\}$ is equal to

(a) $\frac{1}{2}$; (b) 1; (c) -2; (d) -1; (e) 0.

[2]

3. The maximum increase of the function $f(x,y) := e^{x^2-y}\sqrt{y}$ at the point $(1,4)$ is equal to:

(a) $e^{-3}\sqrt{16+\frac{49}{16}}$; (b) $16e^6$; (c) $\frac{3}{2}e^{-3}$; (d) $\frac{7}{4}e^{-3}$; (e) $\sqrt{e^{-3}\Big(16+\frac{49}{16}\Big)}$.

[2]

4. Find the cylindrical coordinates (r,θ,z) of the point $\big(\frac{\sqrt{3}}{4}, \frac{1}{4}, \frac{\sqrt{3}}{5}\big)$.

(a) $\big(\frac{1}{4}, \frac{\pi}{6}, \frac{\sqrt{3}}{5}\big)$; (b) $\big(\frac{1}{4}, \frac{2\pi}{3}, \frac{\sqrt{3}}{5}\big)$; (c) $\big(\frac{1}{2}, \frac{\pi}{6}, \frac{1}{2}\big)$; (d) $\big(\frac{1}{2}, \frac{\pi}{3}, \frac{\sqrt{3}}{5}\big)$; (e) $\big(\frac{1}{2}, \frac{\pi}{6}, \frac{\sqrt{3}}{5}\big)$.

[2]

5. Let $D := \{(x,y) \in \mathbb{R}^2\colon\ x^2+y^2 \leq 1, \quad x \leq 0, \quad x\sqrt{3} \leq y \leq -x\sqrt{3}\}$. If the description in polar coordinates of the domain D is given by $\{(r,\theta)\colon\ 0 \leq r \leq 1, \quad \theta_1 \leq \theta \leq \theta_2\}$ then

(a) $\begin{cases} \theta_1 = -\frac{\pi}{3}, \\ \theta_2 = \frac{\pi}{3}; \end{cases}$ (b) $\begin{cases} \theta_1 = -\frac{\pi}{3}, \\ \theta_2 = \frac{2\pi}{3}; \end{cases}$ (c) $\begin{cases} \theta_1 = \frac{2\pi}{3}, \\ \theta_2 = \frac{4\pi}{3}; \end{cases}$ (d) $\begin{cases} \theta_1 = \frac{2\pi}{3}, \\ \theta_2 = \frac{5\pi}{6}; \end{cases}$ (e) $\begin{cases} \theta_1 = \frac{2\pi}{3}, \\ \theta_2 = \frac{7\pi}{6}. \end{cases}$

[2]